COURSE STRUCTURE FOR B.TECH. Mechanical Engineering SECOND YEAR SEM – III (w. e. f. 2024-25)

SEMESTER-III (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-III											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	PRO	24INT151	Civic and Social Service Internship	0	0	0	1	0	-	-	-	-	-	-	
2	BSC	24MA201T	Mathematics – III	3	1	0	4	4	25	25	50	-	-	100	
3	PCC	24ME201T	Introduction to Artificial Intelligence	3	0	0	3	3	25	25	50	-	-	100	
4	PCC	24ME202T	Engineering Mechanics	3	0	0	3	3	25	25	50	-	-	100	
5	PCC	24ME203T	Thermodynamics	3	0	0	3	3	25	25	50	-	-	100	
6	PCC	24ME203P	Thermodynamics Laboratory	0	0	2	1	2	-	-	-	25	25	50	
7	PCC	24ME204T	Mechanical Measurement and Metrology	3	0	0	3	3	25	25	50	-	-	100	
8	PCC	24ME204P	Mechanical Measurement and Metrology Laboratory	0	0	2	1	2	-	-	-	25	25	50	
9	PCC	24ME205P	Mechanical Drawing Laboratory	0	0	2	1	2	-	-	-	25	25	50	
			Total	**15**	**1**	**6**	**20**	**22**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24MA201T					Mathematics – III					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs. / Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. The primary objective of this course is to provide students with a comprehensive understanding of advanced mathematical transforms and their applications in engineering.
2. To use the concept of Laplace transform and its significance in solving differential equations.
3. Understand the fundamental concepts of difference equation and Z-transform.
4. To use this basic course in upcoming courses in respective specializations in higher classes.

UNIT 1 Laplace Transform **10 Hrs.**

Piecewise continuous functions and exponential functions, Definition, Existence and Properties of Laplace transforms, Heaviside function, Impulse function, Inverse Laplace transform, Properties of inverse Laplace transforms, Convolution theorem, Applications of Laplace Transforms in solving differential equations.

UNIT 2 Difference Equation and Z-Transform **11 Hrs.**

First and second order difference equation with constant coefficient, Solution of difference equation-complementary function-Particular integral, Z - transform, Properties of Z-transforms, Convolution of two sequences, inverse Z-transform, Solution of Difference equations using Z-transform

UNIT 3 Fourier Transform **10 Hrs.**

Fourier Integral, Fourier Cosine and Sine transform, Discrete Fourier Transform, Continuous Fourier Transform, Linearity, time shifting, frequency shifting, and convolution properties, Applications in image processing, communication systems, and quantum mechanics.

UNIT 4 Partial Differential Equations

Formation of Partial Differential Equations (PDEs), Solutions of PDEs of the first order, Cauchy problem for first-order PDEs, Lagrange's method, Charpit and Jacobi methods for solving first-order nonlinear PDEs. Classification of second order PDEs, one-dimensional heat equation, wave equation, and Laplace equation, Applications of PDE's to wave analysis.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

At the end of the course, students will be able to:

- **CO1** : Define the partial differential equations of first and second order in order to model or understand the Engineering applications.
- **CO2** : Understand the techniques of Laplace transforms, Z-transforms, and Fourier transforms to solve critical mathematical problems.
- **CO3** : Apply the methods of Transform techniques to solve differential equations involving piecewise continuous and exponential functions.
- **CO4** : Classify the partial differential equations of different orders and solve those using different techniques.
- **CO5** : Appraise engineering or scientific problems that can be addressed using transform techniques.
- **CO6** : Formulate and create various application problems in engineering using series and Transform techniques.

TEXT/REFERENCE BOOKS

1. R.K. Jain & S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publishing House.
2. E. Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME201T					Introduction to Artificial Intelligence in Mechanical Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	0	0	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the utility of Artificial Intelligence for mechanical engineering applications.
2. To understand the machine learning framework.
3. To formulate and implement machine learning methods for real world applications.
4. To explore advanced AI topics in mechanical engineering.

UNIT I: FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE IN MECHANICAL ENGINEERING **08 Hrs.**

Introduction to Artificial Intelligence, Machine Learning, Deep Learning, Artificial Intelligence Vs Automation, Architecture of AI implementations in Mechanical Engineering, Basics of data preprocessing and feature engineering for mechanical engineering dataset.

UNIT II: MACHINE LEARNING TECHNIQUES IN MECHANICAL ENGINEERING **11 Hrs.**

Supervised, Unsupervised learning and semi supervised learning, Regression and Classification algorithms, Evaluation metrics in ML, Feature selection and extraction techniques, Principal Component Analysis, Singular Value Decomposition.

UNIT III: IMPLEMENTATION OF MACHINE LEARNING MODELS **11Hrs.**

Implementation of Linear Regression, Logistic Regression, Naive Bayes, Linear Discriminant Analysis, K-Nearest Neighbour, Artificial Neural Network, Support Vector Machine. Applications of deep learning in mechanical engineering for classification, regression, image processing and time series data.

UNIT IV: ADVANCED TOPICS IN ARTIFICIAL INTELLIGENCE FOR MECHANICAL ENGINEERING **10Hrs.**

Principles of predictive maintenance using AI techniques, Fault diagnosis methods and algorithms for mechanical systems, AI-driven control and decision-making in robotic systems, Applications of vision system and object detection in real-world applications.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand AI fundamentals and its framework.
CO2 : Implement and Analyze data preprocessing techniques.
CO3 : Interpret classification and regression in mechanical engineering applications.
CO4 : Implement diverse machine learning algorithms for various applications.
CO5 : Apply machine learning for systematic design analysis.
CO6 : Develop comprehensive strategies for integrating AI and ML techniques into mechanical engineering processes.

TEXT/REFERENCE BOOKS

1. E. Alpaydin, "**Introduction to Machine Learning**", Pearson.
2. S. Rogers & M. Girolami, "A First Course in Machine Learning", Chapman & Hall.
3. C. M. Bishop, "**Pattern Recognition and Machine Learning**", Springer.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME202T					Engineering Mechanics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the concept of equilibrium condition of loaded members.
2. To obtain centre of gravity, area and mass moment of inertia of different sections.
3. To get acquainted with the kinetics, kinematics and dynamics of particles and rigid bodies.
4. To get accustomed to concept of work-energy methods and its applications.

UNIT I: FREE BODY DIAGRAMS, EQUILIBRIUM AND TRUSS 10Hrs.

Principles of statics, coplanar, concurrent and non-concurrent, parallel and non-parallel forces, Conditions of Rigid body Equilibrium, Equations of Equilibrium, Two and Three force members, Equilibrium in three dimensionsTruss Simple determinate plane trusses and analysis for member forces using methods of joints and methods of sections. Types of beams and loads, Basics of Shear Force Diagram (SFD) and Bending Moment Diagram (BMD)

UNIT II: DISTRIBUTED FORCES, CENTER OF GRAVITY AND MOMENT OF INERTIA 8Hrs.

Center of gravity of lines, plane areas, volumes and bodies, Pappus – Guldinus theorems, moment of inertia, polar moment of inertia & radius of gyration of areas, parallel & perpendicular axes theorems.

UNIT III: FRICTION AND VIRTUAL WORK 12Hrs.

Theory of friction, static and sliding friction, friction and its applications including rolling friction, belt-pulley, brakes, clutches, screw jack, wedge, vehicles. Principle of Virtual work for a particle and rigid body & System of a connected rigid bodies, Conservative Forces, Potential Energy, potential Energy criterion for equilibrium, Stability of Equilibrium

UNIT IV: KINETICS OF A RIGID BODY 10 Hrs.

Kinematics and dynamics of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy formulations

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Determine equilibrium equations of a body subjected to system of forces.
CO2 : Calculate the location of centroid, moment of inertia, polar moment of inertia of standard sections.
CO3 : Apply the theory of friction to different mechanical applications.
CO4 : Estimate the energy criterion for equilibrium condition.
CO5 : Analyse the kinematics and dynamics of a body undergoing translatory, rotary and rigid motion.
CO6 : Evaluate the linear and angular moment of system of particles and rigid body.

TEXT/REFERENCE BOOKS

1. R.C. Hibbeler, Ashok Gupta **" Engineering Mechanics Statics & Dynamics"**, Pearson Publication
2. Ferdinand P. Beer,E. Russell Johnston Jr. **"Vector Mechanics For Engineers Statics And Dynamics"**, Tata Mcgraw hill publication
3. R. C. Hibbeler, **"Mechanics of Materials SI"**, Pearson Publication
4. Beer and Johnston, **"Mechanics of Materials"**, Tata Mcgraw hill publication
5. S S Bhavikatti, **"Engineering Mechanics"**, New Age Publication

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME203T					Thermodynamics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn and understand the basic principles of thermodynamics systems and its properties, energy exchange, reversible, quasi-static, and irreversible processes.
2. To understand the first and second of laws of thermodynamics, concepts of entropy and exergy and its significance in energy efficiency of thermal systems
3. To develop problem-solving abilities, formulate and solve thermodynamics problems using thermodynamic principles and develop the ability to interpret results.

UNIT I: INTRODUCTION AND PROPERTIES OF PURE SUBSTANCE 11Hrs

Introduction: Role and applications of thermodynamics in engineering, concepts of a system and surrounding, Properties of a system, equilibrium, the zeroth law of thermodynamics, heat and work, various thermodynamic processes, **1st law of thermodynamics:** Concept of energy and various forms of energy; internal energy; enthalpy; specific heats; application of first law to closed and open systems; **Properties of pure substances:** Thermodynamic properties of pure substances in solid, liquid and vapour phases, P-v-T behaviour of simple compressible substances, thermodynamic property tables and charts, ideal and real gases, ideal gas and van der Waals equation of state; law of corresponding states, compressibility factor and generalized compressibility chart, Joule-Thomson coefficient of volume expansion, adiabatic and isothermal compressibility

UNIT II: 2ND LAW OF THERMODYNAMICS, ENTROPY, AND EXERGY 11Hrs

Second law of thermodynamics: Limitations of the first law of Thermodynamics, concept of heat engines and heat pumps/refrigerators, Kelvin-Planck & Clausius statement and their equivalence; reversible and irreversible processes; Carnot cycle and Carnot principles/theorems; Thermodynamic temperature scale, Carnot efficiency; **Entropy**: Calusius inequality and concept of entropy; microscopic interpretation of entropy, the principle of increase of entropy, T-s diagrams; **Exergy**: entropy, second law analysis for a control volume, irreversibility and availability, exergy balance and exergy analysis, exergy analysis of open and closed system

UNIT III: THERMODYNAMIC CYCLES 10Hrs

Compressor: classification of compressors, Comparison of adiabatic, polytropic, and isothermal compression process; isothermal, isentropic, mechanical, and volumetric efficiency of a compressor, multistage compression, applications; **Vapour power cycle:** Carnot vapor cycle, Rankine cycle, comparison of Carnot and Rankine cycle, calculation of cycle efficiencies, variables affecting efficiency of Rankine cycle, reheat cycle, regenerative cycle, reheat-regenerative cycle, feed water heaters; **Gas power cycles:** Recapitulation of Carnot, Otto and Diesel cycle, dual cycle, comparison of Otto, Diesel and dual cycles, air standard efficiency, mean effective pressure, brake thermal efficiency, relative efficiency, ideal Brayton cycle

UNIT IV: GAS MIXTURES AND COMBUSTION 8Hrs

Ideal gas mixtures: Dalton's and Amagat's laws, properties of ideal gas mixtures; simple thermodynamic processes involving gas mixtures; **Combustion:** Introduction, combustion reaction equations and stoichiometric air-fuel ratio, low and high calorific values, bomb calorimeter and Junkers gas calorimeter

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand and use thermodynamic terminology correctly; identify type of systems and various sources of energy involved
CO2 : Apply the first law of thermodynamics and analyse energy interactions across the thermodynamic system and its boundary
CO3 : Evaluate feasibility of thermodynamic processes using concepts of 2nd law of thermodynamics
CO4 : Apply knowledge of entropy and exergy to evaluate the energy conversion processes and systems
CO5 : Apply knowledge of thermodynamic power cycles and refrigeration cycle to solve numerical problems
CO6 : Apply knowledge of combustion stoichiometry to determine air-fuel ratios for various fuels

TEXT/REFERENCE BOOKS

1. Cengel, Y. A. and Bole. M. A., ***"Thermodynamics: An Engineering Approach"***, 8th Edition, McGraw Hill (2014)
2. Nag, P. K., ***"Engineering Thermodynamics"***, 6th Edition, McGraw Hill (2017)
3. Rajput, R. K., ***"Engineering Thermodynamics"***, 5th edition, Laxmi Publications, 2019
4. Rayner, J, ***"Basic Engineering Thermodynamics"***, 5th edition, Pearson (2008)
5. Borgnakke, Claus and Sonntag, Richard, ***"Fundamentals of Thermodynamics"***, 8th edition, John Wiley & Sons, Inc. (2014)

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME203P					Thermodynamics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To gain insight into basic concepts learned in Thermodynamics theory course by performing hands on experiments.
2. To develop essential experimental skills including proper handling of equipment, data collection techniques, measurement accuracy, and data analysis
3. To grasp thermodynamic principles in a tangible way, reinforcing comprehension of these principles beyond theoretical study

LIST OF EXPERIMENTS

1 Construction and working of reciprocating air compressors
2 Performance test on a reciprocating air compressor
3 Measurement of calorific value of a liquid fuel by bomb calorimeter
4 To study working of a single cylinder 4-stroke diesel engine
5 Load test on a four-stroke single cylinder Mahindra diesel engine
6 Heat balance test on a four stroke single cylinder Mahindra diesel engine
7 Measurement of dryness fraction by separating calorimeter, throttling calorimeter, and separating-throttling calorimeter
8 Study of Rankine cycle for steam power plant
9 Study vapour compression refrigeration cycle
10 Exergy analysis of reciprocating air compressor

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Develop skill to perform quality experiments and get knowledge on experimental uncertainties
CO2 : Apply first and second laws of thermodynamics to various thermal systems
CO3 : Evaluate the performance of thermodynamic systems
CO4 : Construct charts and graphs for data representation
CO5 : Prepare good quality lab reports
CO6 : Develop presentation and team work skills

TEXT/REFERENCE BOOKS

1. Thermodynamics Lab Manual, PDEU
2. Cengel, Y A and Bole M A, ***"Thermodynamics: An Engineering Approach"***, 8th Edition, McGraw Hill (2014)
3. Nag, P K, ***"Engineering Thermodynamics"***, 6th Edition, McGraw Hill (2017)
4. Rajput, R K, ***"Engineering Thermodynamics"***, 5th edition, Laxmi Publications, 2019

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME204T					Mechanical Measurements and Metrology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. Introducing and acquainting individuals with various industrial terminologies related to measurement systems and metrology.
2. Exploring diverse techniques for measuring physical quantities like pressure, force, torque, etc.
3. Introducing the principles of quality control and assurance and emphasizing the importance of measurement within it.
4. Demonstrating the use of automated inspection and machine vision techniques for precise measurements.

UNIT I: FUNDAMENTALS OF MEASUREMENT SYSTEMS 10Hrs.

Principal of measurements, metrology: introduction and types, methods of measurements, basic terminology of measurements. Standard of measurement: Roles of standards in measurement, material standards, types of standards, subdivision of standards, calibrations. Transducers for Measurement: Introduction to Transducers, its classification, Quality Attributes

UNIT II: LIMITS, FITS, TOLERANCE, AND LIMIT GAUGING 13Hrs.

Principle of Interchangeability, Selective Assembly, Tolerances, allowance, Maximum and Minimum Metal Conditions, Fits, System of Limits and Fits, Hole basis and shaft basis systems, Design of Limit gauging: Taylor's Principle. Measurement of Force: load cell, proving ring; Measurement of Torque: dynamometer; Pressure Measurement Scales, Ring Balance, Inverted Bell Manometer, Dead-weight Pressure Gauge, Measurement of Vacuum, High-Pressure Measurement. Measurement of Strain: Strain Gauge Material, Backing or Carrier Materials, Adhesives, Protective Coatings, Bonding of Gauges.

UNIT III: LINEAR AND ANGULAR METROLOGY 12Hrs.

Linear measurement instruments: verniers and micrometers; Angle measurement instruments: bevel protector, sine bar and center, clinometers, collimator; Calibration of the instruments, slip gauges. Comparators: Need for comparators, characteristics of comparators, and their classifications. Measurement of surface roughness: terminology of surface texture, symbolic representation, measurement techniques

UNIT IV: QUALITY CONTROL AND MISCELLANEOUS METROLOGY 10Hrs.

Introduction to Inspection and Quality Control, Quality Control and Quality Assurance, Statistical Quality Control. Uncertainty of Measurement-Uncertainty Issues, Uncertainty Calculations, Total Quality Management, Six Sigma, Quality Standards. Precision Instrumentation Based on Laser Principles, Coordinate Measuring Machines, Machine Tool Metrology- Nano-technology instrumentation, Automated Inspection, Machine Vision.

TOTAL HOURS: 45 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

1. Understand the fundamentals of measuring systems, their terminology, and the roles of standards and Transducers
2. Analyze the measurement techniques for force, pressure, torque, strain, speed, and temperature systems.
3. Apply the principles of limits, fits, and tolerance for designing of industrial gauges.
4. Apply the concepts of Linear and angular metrology to study the functioning of different comparators.
5. Appraise the Understand the measurement system for roughness measurement and texture
6. Evaluate the different aspects of quality assurance and control and the role of measurement in it.

TEXT/REFERENCE BOOKS

1. D. S. Kumar, "**Mechanical Measurements and Control"**, Metropolian publisher.
2. R. K. Jain, "**Engineering Metrology"**, Khanna Publisher.
3. A.K. Sawhney, "**Mechanical Measurement and Instrumentation**", Dhanpat Rai Publication.
4. G. T. Smith, "**Machine tool metrology-An industrial handbook**", Springer

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME204P					Mechanical Measurements and Metrology Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To learn the working principles of different measuring devices and their precision.
2. To familiarize with various types of measurements, methods, errors, and their limitations.
3. To develop an ability to validly use measuring instruments.
4. To appreciate the importance of measurements and metrology on the quality of a product/component.

LIST OF EXPERIMENTS

1 To measure the various dimensions of the given workpiece by various measuring Instruments.
2 Introduction to generalized measurement system and Terminology.
3 To study the working of the instruments used for measuring environmental parameters and analysis of environmental parameters at different locations
4 To study Statistical analysis (regression analysis, Linear fit, exponential fit, and polynomial fit) of experimental data using EXCEL.
5 To study and calibrate the precision measuring instruments like Vernier caliper, Micrometre, and Dial gauge.
6 To get acquainted with the sine bar.
7 Surface analysis of various manufacturing processes.
8 To measure the fundamental dimensions of a gear using a contour (profile) projector.
9 To study different types of the comparators.
10 To demonstrate different types of Limit Gauges.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the working principles and operations of various instruments.
CO2 : Analyse the effect of different parameters on the measurements and methodologies.
CO3 : Demonstrate the significance of different instruments for different measurements.
CO4 : Judge the uses of appropriate techniques for different measurement applications in practical life.
CO5 : Appraise the limitations of each measurement technique and the methodologies followed.
CO6 : Evaluate the qualitative significance of the quantitative results obtained.

TEXT/REFERENCE BOOKS

1. Lab manual PDEU
2. R. K. Jain, "**Engineering Metrology"**, Khanna Publisher.
3. A. K. Sawhney, "**Mechanical Measurement and Instrumentation**", Dhanpat Rai Publication.
4. G. T. Smith, "**Machine tool metrology-An industrial handbook**", Springer

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – III

24ME205P					Mechanical Drawing Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To be able to draw computer-aided mechanical drawings of components and
2. To be able to perform assemblies of industrial machines, machinery parts and other mechanical equipment by CAD software

LIST OF EXERCISES:

1. Recapitulate concepts of sketch entities and tools, 3D features
2. Use advanced 3D features like lofted base, revolved base, sweep, draft, wrap, and mirroring features.
3. To create different mechanical elements (Trusses, beams, columns, cylinders, patterns, etc.)
4. To create a 3D assembly of mechanical systems and edit parts in the assembly.
5. Create and edit Bill of Materials (BOM)
6. Use a motion study tool to animate the assembly and detect interference/clearance.
7. To create threads using ANSI standards. To study thread features, such as internal and external threads, threaded blind holes, and thread pitch.
8. Tolerance – Limit tolerance, angular tolerance, and geometric tolerances.
9. Fit Tolerances – Clearance fits, hole basis & shaft basis, Interference fits
10. Basic introduction of bearing toolbox
11. To create gears and gear assembly.
12. To study power transmission from shaft to gears and create bearing-gear assembly
13. Assembly drawings using industry standards

COURSE OUTCOMES

Upon completion of the course, students will be able to

CO1 : Demonstrate competency with multiple drawing and modification commands in SolidWorks.
CO2 : Create three-dimensional solid models.
CO3 : Create three-dimensional assemblies of mechanical systems with permissible tolerances.
CO4 : Apply industry standards in the preparation of technical mechanical drawings.
CO5 : Communicate and present ideas and solutions to design problems using an interactive motion study tool.
CO6 : Design/draft the next innovative thing

TEXT/REFERENCE BOOKS

1. Zeid, Ibrahim, "**Mastering Solidworks**". Prentice Hall Press, 2010.
2. Weber, Matt, and Gaurav Verma, "**SolidWorks 2015 Black Book**". CreateSpace Independent Publishing Platform, 2014.
3. Rao, Posinasetti Nageswara, "**CAD/CAM: principles and applications**". Tata McGraw-Hill Education, 2004